Same Weights, Three Ways to Answer

Why Collective Reasoning Needs More Than Communication

Oct 9, 2026 · @Artem Zvyagintsev

Asked to pick one option in a single pass, a frontier model cannot check a two-step rule (59% on pairs, near chance) and follows a false rule it is told 76% of the time. Let the same model think for a few hundred tokens and it checks almost perfectly (100%) and follows the false rule only 1% of the time. Without thinking, switching APIs changes nothing. On a grid of 100 copies, each with too little evidence to find the rule alone, the right two-step rule spreads only when agents can check what their neighbours tell them, by reasoning or with an external check. An update at the end shows that this works even when no agent is given the answer.

The setup

The task is to check a rule against evidence. A hidden transformation maps 4-symbol strings over A–H to other strings, and the model sees a few input → output examples. The catalog has 101 candidate rules: 17 single operations (`reverse`, `rotate_left(k)`, `shift_symbols(k)`, `swap(i,j)`) and 84 compositions of two, such as `shift_symbols(1);swap(1,2)`.

Three tests, all on held-out rules:

1. **Pairs.** One true and one false candidate rule, named; does the model prefer the true one? 200 pairs, 100 single-op and 100 two-op, plus 100 three-op pairs from a separate held-out set of compositions outside the catalog. Chance is 50%.
2. **Told a lie.** A message names a false rule as the answer; does the model follow it, or recover the true rule from its examples? 84 two-op cases.
3. **Pragmatic listening.** A reference game where the right answer depends on what the speaker would have said instead (46 instances, 153 words). Scored against an ideal pragmatic listener.

One model, `gpt-6-luna` (OpenAI, public beta), asked three ways:

- **Decisions API:** one choice question, the model returns a probability for every option in one pass. Averaged over 4 option orderings.
- **Responses API, reasoning off** (`effort=none`): the model writes the option key directly.
- **Responses API, reasoning on** (`effort=medium` and `high`): the model may think before writing the key.

Responses modes were sampled 8 times per question across the 4 orderings; all modes are compared on the chosen answer, not on probabilities. Predictions and pass thresholds were committed before any Responses number existed (tag `prereg-luna-modes-lock`; see PROVENANCE.md). The two that matter most: on two-op pairs, reasoning high beats Decisions by at least +0.15 (primary); with reasoning off, the two APIs differ by less than 0.10 (control).

Results

[embedded content: luna_modes.json · gpt-6-luna public beta, Oct 2026 · pairs n=100 two-op, lie n=84 two-op, pragmatic 46 instances / 153 words]

Paired cluster-bootstrap intervals against Decisions: reasoning high +0.41 [+0.32, +0.50] on two-op pairs, reasoning off +0.01 [−0.06, +0.08], and +0.35 [+0.27, +0.44] on pragmatic listening. On three-op pairs the gap widens: reasoning medium +0.475 [+0.35, +0.59] over Decisions, which falls to 52%. Medium and high sit within two points of each other wherever both ran, so depth, not effort level, is what the budget has to cover: mean reasoning tokens grow from 227 (one op) to 689 (two) to 1,349 (three).

What it means

The model has the components the task needs; in one pass, it fails to compose them. Checking a two-op rule means applying the first operation, holding the intermediate string, applying the second and comparing. In a single pass that intermediate result has to be computed inside the network, with nowhere to write it down, and here the model does not manage it. A few hundred tokens of reasoning give it a place to write. This is a limit of the mode as observed, not a proof that one pass cannot compose: whether training can teach a one-pass model to do this step internally, in a way that transfers to compositions and environments it was not trained on, is an open question and the one I want to test next.

Three observations point the same way:

- **It is a cliff, not a slope.** One pass scores 94% on single operations, 59% on two and 52% on three: it does not degrade with depth, it drops to chance at the first composition. Reasoning stays at the ceiling at all three depths while its token count roughly doubles per added step.
- **The control holds.** Reasoning off, Responses − Decisions on two-op pairs is +0.01 [−0.06, +0.08]. The gain comes from thinking, not from a different API.

- **An external step works as well as an internal one.** When the environment attaches a ready-made ✓/✗ check to each candidate rule, the same one-pass model goes from 65% to 99.5% on picking the rule from the full catalog. On pragmatic listening, an explicit step built from the same one-pass model also works: asking it, for each candidate item, what word it would have said, and picking the item that best explains the word heard, lifts it from 61% to 93% [+0.23, +0.40] with no reasoning at all, matching reasoning medium (93%; `pragmatic_argmax_L2` = 0.935 in `luna_modes.json`). Inverting a neural speaker to get a pragmatic listener is an established method (Andreas & Klein, 2016; Fried et al., 2018); what is new here is that it matches reasoning on the same weights. Since the reference listener is itself defined by inverting a speaker, this shows the model's speaker carries what its one-pass listener ignores, not that the model becomes pragmatic in general. Someone has to do the intermediate step; it does not matter much who.

This is the familiar compositionality gap of language models: each step known, the composition not done without an explicit chain. The new part is seeing it inside one set of weights, with the reasoning budget as the only thing changed, and a dose curve that saturates early: medium (mean \~690 reasoning tokens on two-op pairs, \~460 across all pairs) is already at the ceiling.

Caveats

- **One model, one task family.** String transformations over a 101-rule catalog. Other domains may need more or fewer steps. The three-op rules come from a separate held-out set outside the catalog, with two extra options per question, so the 1 → 2 → 3 comparison is not a pure change of depth.
- **Public beta.** `gpt-6-luna` and the Decisions API may change before release. How Decisions mode is built internally is not documented, so I make no claim about it beyond what it outputs.
- **Different estimators.** Decisions returns probabilities; Responses modes are estimated from 8 samples, with ties scored as half. That is why every comparison is on the chosen answer.
- **Small held-out sets.** Intervals are cluster bootstraps over the 26 held-out two-op functions (pairs) or the 46 instances (pragmatic listening).
- **Validity.** 0 invalid answers with reasoning off or medium; 1 truncated answer out of 1,600 at high, where reasoning hit the 8,192-token cap.

Why this matters for multi-agent work

I study whether a population of decision-model agents on a 10×10 grid can spread a correct rule by messaging each other. An agent that follows whatever its neighbours say spreads errors as easily as truths; spreading the truth needs agents that check claims against their own evidence.

These results say one-pass decision models, small or frontier, do not check two-step claims, while the same weights with a little reasoning do. That splits the next steps into two routes:

- **Verify from outside:** the environment attaches checks to messages (our engine-check stand).
- **Verify from inside:** agents that reason, or decision models trained to make the intermediate step within one pass.

I put both on the same grid, with each agent given too little evidence to find the rule alone.

On the grid

> A full account of where this work led is in What None of Them Knew.
>
> The grid runs, round by round for each model, are in Wandering Holes.
>
> **Correction, Oct 13, 2026.** The no-channel control in this section removed two things, not one: the neighbours' messages and each agent's memory of its own previous rule. With that memory restored and still no messages, a single reasoning agent reaches 0.89 on two-step rules (end of run), against 0.99 with messages. The claims below that two-step rules are solved "only collectively" and that messages make them "possible" do not hold for reasoning agents. The single-agent results above are unaffected. See the update at the end.

Single rules are solved by each reasoning agent alone. Two-step rules are solved only collectively, and only when incoming claims are checked: by the agent's own reasoning or by the environment.

The same `gpt-6-luna`, now as 100 cells on a 10×10 torus. Four source cells hold the true rule. (For reasoning cells they turn out not to matter; see the update.) Every other cell sees one example that fits more than one rule, so it cannot identify the rule alone. Each generation a cell picks a rule, is told whether it was right and, with the channel on, publishes its rule to its neighbours. J is the share of non-source cells holding the true rule, averaged over 10 generations; 5 seeds per depth. "Solved" below means most non-source cells come to hold the true rule and keep holding it, not that some cells find it.

cells	channel	check on incoming rules	J, one-op	J, two-op
reasoning medium	on	own reasoning	0.99	0.82
reasoning medium	off	—	0.93	0.25
Decisions (argmax)	on	none	0.53	0.14

cells	channel	check on incoming rules	J, one-op	J, two-op
Decisions (argmax)	off	—	0.19	0.00
Decisions (argmax)	on	engine ✓/✗	0.88	0.67

[image: J by generation: reasoning and one-pass cells, with and without the channel, and one-pass cells with engine checks; one-op and two-op panels]

Share of non-source cells holding the true rule, by generation; bands are 95% bootstrap intervals over 5 seeds. "Silent" = no channel; "one-pass" = Decisions API; "+ check" = engine ✓/✗ on incoming rules. The dashed black line is the topological reachability bound: how fast the rule could spread on this grid if every cell adopted it on first contact.

- **Messages make two-op rules possible, not just faster.** With the channel, reasoning cells lock on the true rule in all 5 two-op seeds by generation 2–4; without it they stay at 0.25 (+0.57 [+0.55, +0.61]). On one-op rules each reasoning cell finds the rule alone, so the channel adds little.
- **Collective correction has two stages.** A reasoning cell checks each incoming rule against its own example and does not adopt one that contradicts it (98% of the time, in the three seeds traced). Among the rules that fit, it switches to the true one when told its own answer was wrong and the true rule is in its inbox (87%). One-pass cells keep the second stage and lack the first: in the three seeds I traced, all 70 cells that switched to a neighbour's rule had just been told they were wrong, but they took rules contradicting their own example as readily as rules that fit it. That stage alone still lets some truth through: without the channel one-pass cells never reach a two-op rule (J 0.00); with it they reach 0.14 (+0.13 [+0.03, +0.25]). I had predicted no difference. On one-op rules, which one-pass cells can check, the channel matters more: 0.53 against 0.19 (+0.34 [+0.20, +0.47]).
- **Identical agents agree without talking, sometimes on the wrong answer.** In one two-op task, the first operation of the true rule alone fits 97% of the cells' examples. Without a channel, 78% of reasoning cells picked that simpler rule independently in the first generation (31% would pick a single-op rule if cells chose evenly among the rules that fit their example) and held it for all ten, although they were told their answer was wrong in 90% of generations: with nothing better in sight, "wrong" alone did not move them. Knowing an answer is wrong is not the same as having a better one, even for a reasoning model. One-pass cells did the same (98%). With messages, reasoning cells left it within a generation. A shared prior makes copies of one model agree; agreement alone is no evidence that they communicated. This is one task out of five, so it is an observation, not a rate.
- **Delegated verification works for one-pass agents.** With an engine ✓/✗ attached to each incoming rule, one-pass cells reach 0.67 on two-op rules (0.92 by generation 10), against 0.14 without (+0.53 [+0.31, +0.70]). It helps on one-op rules too, 0.88 against 0.53 (+0.35 [+0.16, +0.56]): picking from 101 rules with one example is harder than choosing between two. On two-op rules it is slower than reasoning cells, and not yet level with them: by generation 10, 0.92 against 0.99.
- **An engine check cannot break that kind of agreement.** The check tests whether a rule fits the examples, and the simpler false rule does: it solves local consistency, not identifiability. In that task, one-pass cells with engine checks locked on the false rule for 7 of 10 generations before recovering. Only the combination of "you were wrong" and the true rule arriving from a neighbour breaks it.

Caveats specific to the grid: one model, 5 seeds per depth, one 10-generation epoch, and a lock threshold (modal share ≥ 0.5) chosen in advance; at 0.6 the false-agreement result above survives and two borderline cases disappear.

The pattern is the one from the single-agent tests, scaled up: in these experiments, collective inference worked only when incoming claims were checked, by the agents or by the environment, and a channel without a check passed errors along with truths. Whether a one-pass model can be trained to make that check itself is the next question.

Update, Oct 11, 2026: does the grid need a teacher?

The grid runs above seed four "source" cells with the true rule. I reran the two-op tasks with one source and with none: same model, same five tasks, same private examples, with predictions committed before the runs.

reasoning cells, two-op	J at generation 10	J averaged over 10 generations
4 sources (as above)	0.99	0.82
1 source	0.98	0.80
0 sources	0.99	0.82
0 sources, no messages	0.25	0.25

No teacher needed. With no seeded source, all five populations converge on the true rule (98–100% of cells by generation 10). In the first generation, 23–39% of cells find the two-step rule on their own, as they do without messages. The rest are told they are wrong and check the candidates their neighbours send. Even with one source, 70–88% of the cells holding the rule at the end trace it only to such solo finds, not to the source.

A minority overturns a majority. In the task where the simpler first operation fits almost every example, 75% of cells start on the false rule and 23% on the true one. Without messages, the false majority holds for all ten generations. With messages and no source, the true rule is the majority by generation 3 and is held by 98% of cells at generation 10. I had predicted the opposite: that without a seeded source the population would stay locked on the false rule.

Checks narrow the field; votes break ties. A cell that has been told it is wrong, and has only one incoming rule that survives its own check, usually switches on a single neighbour's message (77% per generation; 87% with three or more

neighbours sending it). When two rules survive the check, the number of neighbours decides (25% with one, 75% with three or more). One-pass Decisions cells follow the crowd either way (4% with one, 27% with three or more): they have no check to narrow the field.

One-pass decision cells behave differently. They almost never find the rule alone, so they depend on the source. With an external check on incoming claims, a single source's rule travels the whole grid (92% of cells at generation 10, 80% tracing it back to the source). Without the check, the wave dies within a few steps (10%).

> *Oct 13: with engine checks, one-pass populations also reach the rule without a source (0.79 at generation 10, all five tasks); see below.*

What this changes. The conclusion above stands: collective inference needed a check on incoming claims. But "a seeded truth spreading through the grid" is the wrong picture for reasoning cells. In these runs the true rule was present from the first generation, so what the grid did was select among hypotheses some agents already held. In follow-up runs where each agent's evidence was more ambiguous, the true rule sometimes won although no agent held it at the start; I am testing what drives that.

Caveats: one model, five tasks, one 10×10 grid on which no cell is more than five steps from any other. The share of solo finds depends on how much evidence each cell gets. With more ambiguous evidence the outcome varies: at the highest ambiguity I tried, three of five populations ended on a false rule. Two of my predictions for these runs failed: I expected most switches on the wavefront to involve a single neighbour (19% did, because solo finds are everywhere), and I expected the zero-source population to stay on the false rule in the hard task (it did not).

Edited Oct 12: two sentences narrowed after follow-up runs.

Edited Oct 13: one-pass zero-source line.

Update, Oct 13, 2026: it was the memory

One line in the prompt

Each generation, a cell sees one line about its previous step. With the channel on, a cell that published its rule sees it:

```
Previous step: answer=BCBH correct=false published signal=0 say='shift_symbols(1);swap(2,3)'.
```

With the channel off, the same cell saw this:

```
Previous step: answer=BCBH correct=false silence.
```

It knew it had been wrong, but not which rule had been wrong. The memory code stored the rule only as the content of the message, so turning messages off also erased the agent's record of its own hypothesis. Every "no channel" control in this article changed two things at once.

The test

I added the rule back to the silent agents' memory line and changed nothing else:

```
Previous step: answer=BCBH correct=false silence held_rule='shift_symbols(1);swap(2,3)'.
```

A line-by-line diff of the rendered prompts confirmed this was the only difference, and the request hashes in the API cache confirmed the model received it. Predictions were committed before the runs. Same five two-step tasks, same examples, ten generations.

reasoning cells, two-step	J at generation 10	J averaged over 10 generations
messages (as above)	0.99	0.82
no messages, own previous rule shown	**0.89**	**0.76**
no messages, own previous rule hidden (as above)	0.25	0.26

The agent with its own rule in memory beat the agent without it in all five tasks. I had predicted that messages would still account for at least 0.20 of the gap; they account for 0.10.

Why memory matters this much

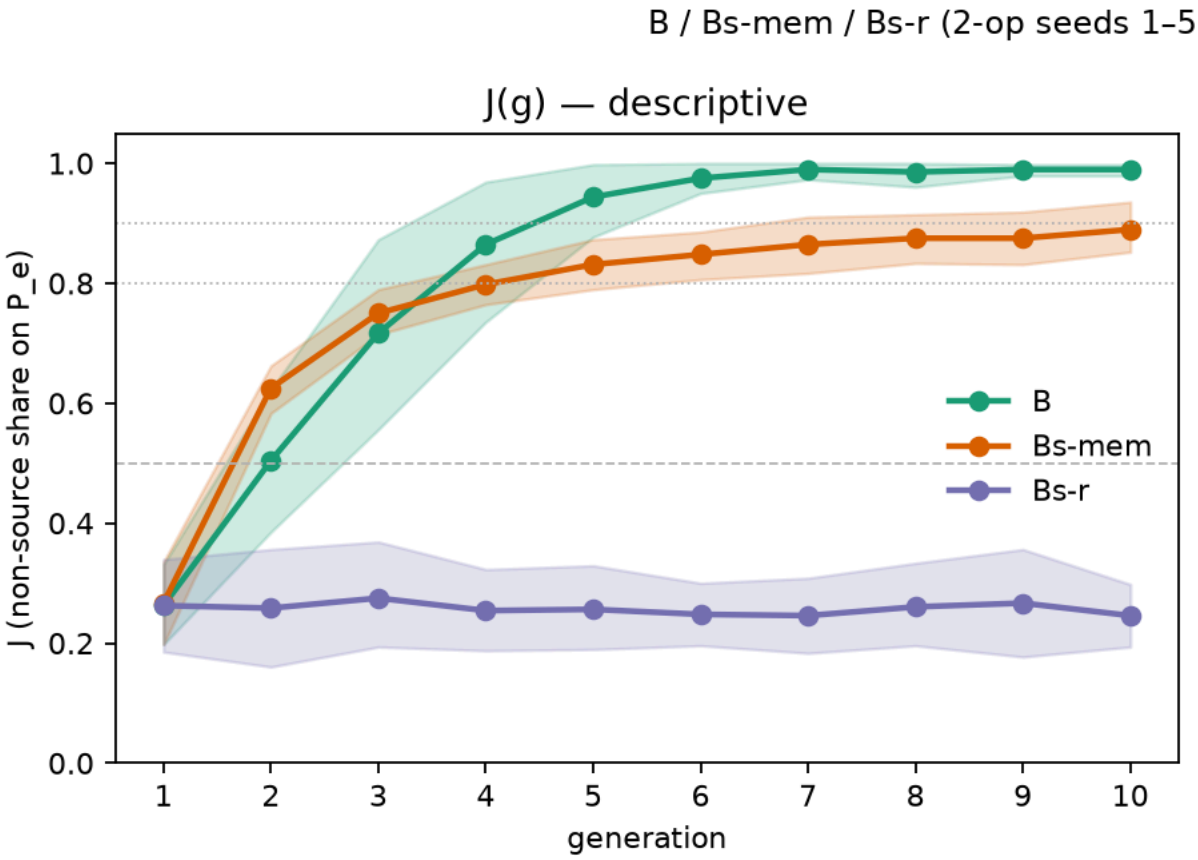
Memory works in two ways.

- **Finding.** Told it was wrong, an agent that sees its refuted rule tries something else, and finds the true rule three times as often (0.31 vs 0.11 per generation). An agent that does not see it re-derives the same rule from the same example and repeats the mistake: it kept the same rule after "wrong" 94% of the time, against 8% with memory. The earlier observation that "'wrong' alone did not move them" was this. The agents could not tell which answer had been wrong.
- **Keeping.** An agent that found the true rule and was told it was right kept it 99.7% of the time with memory, and 69% without. Without memory, a correct answer survives about two generations.

These two rates explain the flat line of the memoryless population. If a cell finds the rule with probability h and loses it with probability 1 − r, the steady share is h / (h + 1 − r) = 0.11 / (0.11 + 0.31) ≈ 0.26, against the observed 0.25. Those agents were not failing to solve the task. They kept finding the rule and losing it again.

What messages add

The curves by generation give a different picture from the one above.



- **Not a faster start.** Agents with memory and no messages are ahead early (0.62 vs 0.50 at generation 2). Incoming messages full of the common wrong rule seem to slow independent search, though this is descriptive.
- **The tail.** Alone, 85–90% of cells get there and stop. 0.9 was reached in two of five tasks. With messages, all five tasks pass 0.9 by generation 3–6. The cells that never find the rule alone are not stuck on a lucky wrong answer. They cycle: two thirds of their generations return to a rule already refuted, often alternating between two (median cycle of 4 returns). A one-step memory cannot rule out what it no longer remembers. With messages, the true rule arrives from a neighbour right after a "wrong": 25 of 27 late switches happened that way.
- **Not cheaper.** With messages, reasoning cells spend more reasoning tokens per generation, not fewer.

At higher ambiguity

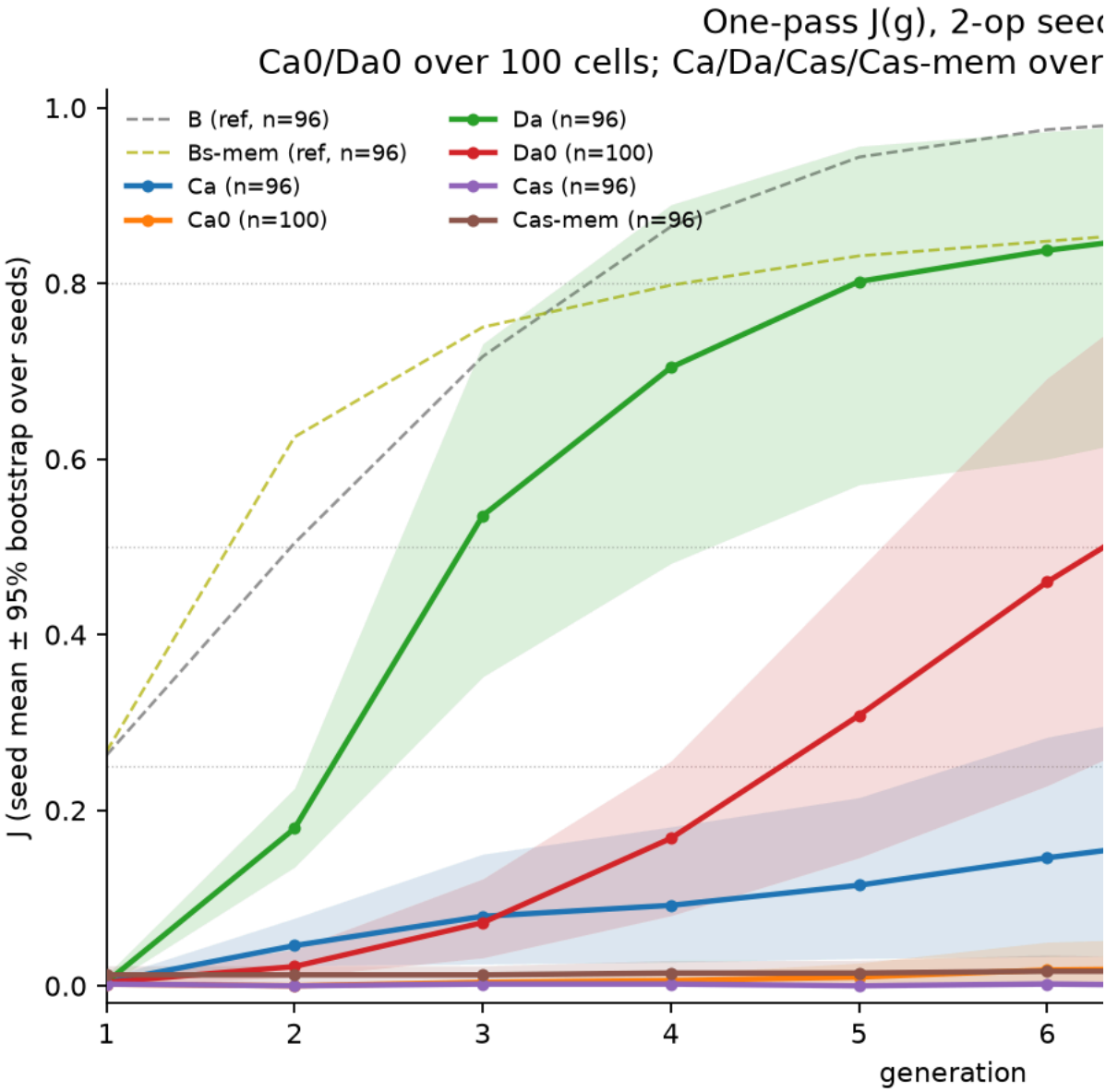
In the harder tasks, where the first operation alone fits almost every example, the result repeats. With its own rule in memory and no messages, an agent proposes the true rule on its own at a rate of 0.039 per generation; with the rule hidden, never. Shown as "published" or not, the rate is the same (0.041 vs 0.039): the rule in memory matters, not publication. These populations also lock on the wrong majority rule without any channel (30–50% of generations), so false agreement there is a shared prior, not a social effect.

One-pass agents: where the collective does real work

One-pass Decisions agents gain nothing from memory on two-step rules (0.02 at generation 10 with their rule shown, 0.00 without). Alone, a one-pass agent hits the true rule by luck in well under 1% of generations. I reran the one-pass populations with no seeded source, with and without engine checks on incoming rules. Predictions were committed before the runs.

one-pass cells, two-step	J at generation 10
alone (no channel)	0.00
alone, own previous rule shown	0.02
channel, no check, no source	0.04

one-pass cells, two-step	J at generation 10
channel, no check, 4 sources	0.25
channel + engine check, no source	**0.79**
channel + engine check, 4 sources	0.92



Without a source and with checks, one-pass populations reached the true rule in all five tasks (0.48 to 1.00 at generation 10), and the curves were still rising. I had predicted that they would stay at least 0.20 below the seeded runs. They came within 0.13. The update of Oct 11 said one-pass cells "depend on the source". With engine checks, they don't.

The mechanism is simple and fully collective. A lucky find appears somewhere. Its neighbours receive it with a ✓, and one checked message converts a neighbour 40% of the time, against 3% without the check. Retention is high either way (about 0.97), so the check works on adoption, not on memory. No member of this population can identify the rule. The population turns a fraction of a percent of individual luck into 79% of cells holding the rule. This is the first effect in these experiments that no single agent, with or without memory, reproduces. It depends on an external check: the environment does the verifying, so it is closer to a search with verified propagation than to agents reasoning together. Whether an agent can take the checker's place is the next test.

Without the check, the outcome is a lottery. A projection from the measured adoption and retention rates (a model, not a run) puts seeded populations without checks anywhere between 0 and 0.94 by generation 30, depending on the task. Such a projection extrapolates average rates; it cannot capture rare ignition events, where one lucky find spreads and the curve bends upward, so the real spread between tasks may be wider. In one task, a cell adopted a correct rule from a neighbour 0.3% of the time. That is not a dislike of the rule: shown only its own example and a choice between the true rule and the common wrong one, the same agent picks the true rule 54% of the time, no less often than in the other tasks. The rule

loses in the full context, among the neighbours' messages and the whole catalog. Without a check, a one-pass agent adopts what looks plausible to it in that context, not what is true.

A third kind of agent sits in between: the reasoning model with reasoning switched off, answering in a short line of text. Alone, it looks like a one-pass agent. On the grid (channel, four sources, no check) it reached 0.47 at generation 10, against 0.25 for one-pass agents and 0.99 with reasoning. I had predicted it would match the one-pass agents; it did not. In the task where one-pass populations stall, it reached 0.56. The raw numbers suggested it checks incoming rules against its own example (it adopted a contradicting rule in 10% of adoptions, against 62% for one-pass agents). Compared in identical inbox situations, it does not: it adopts contradicting rules more often than one-pass agents, and keeps the true rule less often. The raw gap came from what its neighbours were sending. What differs is how easily it changes its mind, in both directions. It picks up the true rule faster and drops it faster: told it was right, it kept the rule 71% of the time, against 97% for one-pass agents. Such a population settles within ten generations; the stickier one-pass population is still moving. Ten generations favour the first; longer runs may not. The advantage also needs the four seeded cells that publish the true rule every generation and never change it: without them, the same population reached 0.01.

This mobility is not sampling noise. At temperature 0 the same agent kept the rule 76% of the time and the population reached 0.62, higher than with default sampling. Making the one-pass agent random instead, by sampling its choice from its own probabilities, did not reproduce the effect: it dropped the rule almost at once (it kept it 9% of the time) and reached 0.01. The reasoning-off agent simply weighs its neighbours more and its own previous answer less. With permanent sources of the true rule nearby, that is enough.

A note on cost

At OpenAI's list prices, the one-pass populations here cost more per task than the reasoning populations: about $1.0–1.1 against $0.33–0.44. The reasoning model with reasoning switched off was the cheapest, about $0.22 per task, with around 28 output tokens per answer. A one-pass answer is averaged over four option orderings, and that matters: with a single ordering, 71% of a lone one-pass agent's rule choices change. That is an API price, not a compute cost. A one-pass decision reads the prompt and scores the options without generating text. On local hardware, four orderings mostly share the same prefix, while a reasoning agent generates hundreds of tokens per answer. On one's own GPU, a slow population of cheap verified agents may well be the economical option. I have not measured that yet.

What still holds

The single-agent results at the top of this article stand: they involve no memory, no generations and no channel. On the grid, the checking behaviour stands. Reasoning agents reject incoming rules that contradict their own example, and engine checks let one-pass agents reach 0.67. The numbers in the tables above are correct. What changes is what they mean.

What I got wrong, and how I found it

The control was mine, and so was the mistake: I checked that the intended input changed, not that nothing else did. A line-by-line diff of two prompts would have shown it on the first day. Every comparison I report now starts with that diff, and the gate fails on any difference not listed in advance.

Four of my post-hoc readings during this work were also overturned by the registered test that followed: - a split inside the message runs suggested discovery did not depend on memory; - I expected the stuck agents to be lucky with a wrong rule; - I called false agreement the one genuinely collective effect; - I read the reasoning-off agents' advantage as a check against their own example.

Several registered predictions failed as well. I expected one-pass agents with checks to need a seeded source. I expected the reasoning-off agent to behave like the one-pass agent on the grid, then to owe its advantage to a check against its own example, then to sampling noise. Each was tested before I wrote it down as a finding, and each was wrong.

A neighbour-based heuristic for "who published" produced a threefold effect that disappeared when I recovered the exact status from the cache keys.

Where this leaves the question

In this task, with this model, the answer depends on the agents.

- **Agents that reason and remember their own attempts** solve the task alone. For them the collective is a delivery system: messages bring the last 10–15% of agents along.
- **Agents that cannot solve it alone** (one pass, no reasoning) need the collective and a check. Together they reach a rule none of them can find, without any teacher. Without the check they depend on luck and on how plausible the right answer looks.

Across all of this, an explicit step does the work: reasoning, remembering one's refuted attempts, or a check on incoming claims. Communication matters when it carries the output of such a step to agents that cannot take it themselves. This fits what others have found when the memory of earlier attempts, including failed ones, is kept explicitly: GitSwarm's shared repository keeps failed work, and the designer in SPADE relies on an accumulated memory of environments.

It also means this task cannot show emergence. Each cell's own example, plus "right/wrong" feedback and ten generations of memory, is enough to identify the rule. A collective is necessary only when no single agent's information is enough: the

evidence is split across agents, or there is no individual feedback to learn from. That is the next experiment, with `gpt-6-luna` as the reference and our own decision models alongside it.

Caveats: one model, five tasks per ambiguity level, a closed catalog of 101 rules, one 10×10 grid. The memory line holds one step. A fuller history of refuted rules may close the remaining gap without messages, which is a test I have not run.

Update added Oct 13, 2026. Prompts, gate diffs, predictions and analyses: `TASK_LIFE_OWN_MEMORY.md`, `prompts/own_memory_gate_diffs.md`, `runs/life_own_memory/`; one-pass arms: `runs/life_onepass_no_source/`; reasoning-off: `TASK_LIFE_LUNA_NONE.md`.